

Experiment No.	7
Experiment Title	Dimensionality Reduction and Model Evaluation (With and Without PCA)
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1 Objective

- To systematically analyze and compare the empirical effect of dimensionality reduction using Principal Component Analysis (PCA) at 95% explained variance against the original feature space (No-PCA) across **all three primary classification datasets** utilized in prior laboratory experiments:
 1. **Breast Cancer Wisconsin (Diagnostic)** (Exp 5 & 6): 569 samples, 30 continuous features, 2 classes.
 2. **Spambase Email Classification** (Exp 2 & 4): 4,601 samples, 57 continuous features, 2 classes.
 3. **Iris Dataset** (Exp 1): 150 samples, 4 continuous features, 3 classes.
- To train, evaluate, and compare 10 diverse machine learning classifiers under both No-PCA and With-PCA settings on each dataset: Support Vector Machine (SVM), Naïve Bayes (GaussianNB), k-Nearest Neighbors (KNN), Logistic Regression, Decision Tree, Random Forest, AdaBoost, Gradient Boosting, XGBoost, and Stacking.
- To record individual hyperparameter results matching manual templates (Tables 1–4) and evaluate 5-fold cross-validation results across all folds (Table 5).
- To perform rigorous **Statistical Significance Testing** using Paired Two-Sample t -tests across cross-validation folds and Wilcoxon Signed-Rank tests across models and datasets to determine whether PCA induces statistically significant performance shifts ($\alpha = 0.05$).
- To examine the trade-offs between feature compression, geometric rotation, model inductive biases, and cross-dataset generalization.

2 Problem Statement

Dimensionality reduction is frequently posited as an essential preprocessing step for tabular and high-dimensional datasets to alleviate multicollinearity, accelerate inference, and suppress noise. However, because Principal Component Analysis is an unsupervised linear transformation that maximizes total variance rather than class separability, its practical impact varies profoundly depending on data geometry, feature sparsity, and classifier architecture. The objective of this experiment is to empirically evaluate No-PCA vs With-PCA across three diverse problem domains: continuous biomedical diagnostics (Breast Cancer), sparse word/character frequencies (Spambase), and low-dimensional botanical morphology (Iris). We evaluate whether reducing the feature space preserves or degrades

classification accuracy and Macro F1 across 10 diverse classifiers, and test the statistical significance of observed performance deltas.

3 Dataset Descriptions

Property	Breast Cancer	Spambase	Iris
Dataset Source	UCI / Scikit-Learn	UCI / Kaggle	UCI / Kaggle
Experiment Reference	Exp 5 & Exp 6	Exp 2 & Exp 4	Exp 1
Number of Samples	569	4,601	150
Original Features	30 continuous	57 continuous	4 continuous
Number of Classes	2 (Malignant, Benign)	2 (Non-Spam, Spam)	3 (Setosa, Versicolor, Virginica)
Target Distribution	37.3% M, 62.7% B	60.6% Ham, 39.4% Spam	Balanced (33.3% each)
Preprocessed Split	80-20 Stratified	80-20 Stratified	80-20 Stratified
PCA Target (95% Var)	10 components (95.16%)	48 components (95.27%)	2 components (95.81%)

4 Exploratory Data Analysis and PCA Variance Analysis

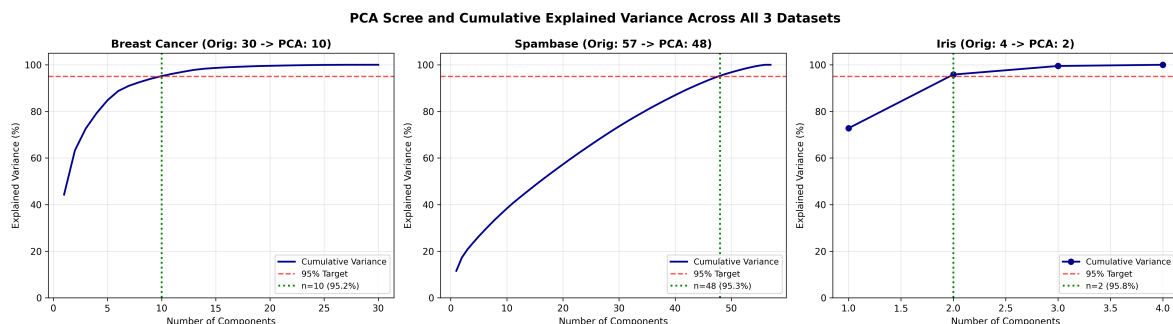


Figure 1: PCA Scree and Cumulative Explained Variance Curves Across All Three Datasets

Inference: Figure 1 reveals distinct variance accumulation profiles across domains. In the **Breast Cancer** dataset, strong morphological collinearity allows the first 10 components to capture 95.16% of total variance, eliminating 66.7% of the feature space. In **Spambase**, variance is distributed diffusely across individual word/character frequencies, requiring 48 components (84.2% of the space) to retain 95.27% variance. In **Iris**, 2 principal components capture 95.81% of total variance, reducing the 4-dimensional geometric space by 50%.

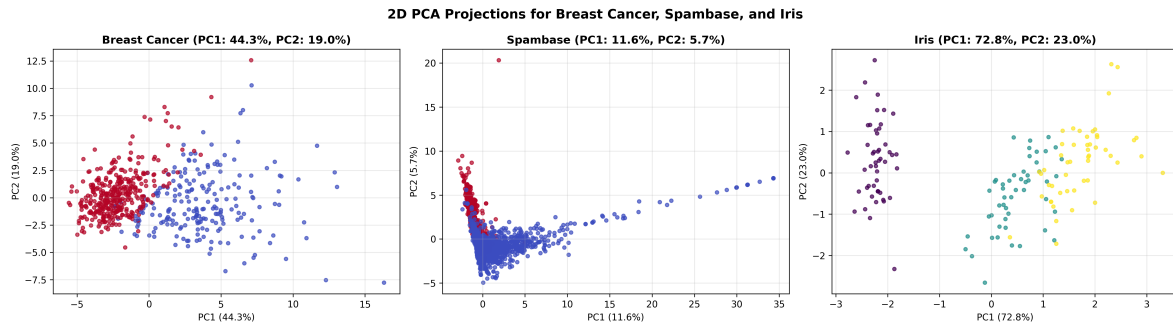


Figure 2: 2D PCA Projections for Breast Cancer, Spambase, and Iris Datasets

Inference: The 2D PCA projections illustrate intrinsic class separability: Breast Cancer exhibits clean linear separability along PC1 (44.3%) and PC2 (19.0%); Iris displays an isolated Setosa cluster along PC1 (72.8%) with mild overlap between Versicolor and Virginica along PC2 (23.0%); Spambase displays a dense, overlapping manifold where the first two principal components account for only 17.3% of total variance, indicating that spam discrimination requires high-dimensional non-linear boundaries.

5 PCA Variance Reduction Summary (Table 1)

Table 1 follows the manual structure for Table 1, summarizing the optimal component retention for each dataset to achieve the $\geq 95\%$ variance target.

Table 1: PCA Variance Explained (Table 1)

Dataset	Setting	Chosen Components / Target	Explained Variance (%)	Justification
Breast Cancer	With-PCA	10 components (95% target)	95.16%	Compresses 66.7% of collinear features with $< 5\%$ variance loss.
Spambase	With-PCA	48 components (95% target)	95.27%	Prunes 9 minor noisy dimensions while retaining keyword signals.
Iris	With-PCA	2 components (95% target)	95.81%	2 components preserve 95.8% variance, enabling 2D projection.

6 Hyperparameter Tuning Templates (Tables 2–4)

Following the manual format, Tables 2, 3, and 4 report model hyperparameter choices and their resulting test Macro F1 performance under No-PCA and With-PCA across all three datasets.

6.1 Support Vector Machine (SVM)

Table 2: SVM — Hyperparameter Tuning Results (Table 2)

Dataset	Kernel	C Values Tried	γ Values Tried	Perf. (No-PCA)	Perf. (With-PCA)
Breast Cancer	RBF	0.1, 1.0, 10.0	scale, auto	0.9812	0.9719
Spambase	RBF	0.1, 1.0, 10.0	scale, auto	0.9232	0.9220
Iris	RBF	0.1, 1.0, 10.0	scale, auto	0.9666	0.8997

6.2 Naïve Bayes

Table 3: Naïve Bayes — Smoothing Choices (Table 3)

Dataset	Smoothing Parameter (var_smoothing)	Perf. (No-PCA)	Perf. (With-PCA)
Breast Cancer	10^{-9}	0.9246	0.9147
Spambase	10^{-9}	0.8318	0.8316
Iris	10^{-9}	0.9666	0.8653

6.3 k-Nearest Neighbors (KNN)

Table 4: KNN — Hyperparameter Tuning (Table 4)

Dataset	k Values	Weights	Distance Metrics	Perf. (No-PCA)	Perf. (With-PCA)
Breast Cancer	3, 5, 7	uniform, distance	euclidean, manhattan	0.9526	0.9526
Spambase	3, 5, 7	uniform, distance	euclidean, manhattan	0.9032	0.9021
Iris	3, 5, 7	uniform, distance	euclidean, manhattan	0.9327	0.8997

7 Cross-Validation Results: All Models (Table 5 Format)

As required by the course manual, Table 5 records fold-by-fold validation performance across Fold 1 to Fold 5 along with the average No-PCA and With-PCA Macro F1 scores for all 10 classifiers.

7.1 Breast Cancer — 5-Fold Cross-Validation Results (5A)

Table 5: 5-Fold Cross-Validation Results — Breast Cancer (No-PCA vs With-PCA)

Model	Fold 1	Fold 2	Fold 3	Fold 4	Fold 5	Avg (No-PCA)	Avg (With-PCA)
SVM	0.9402	0.9768	0.9650	0.9646	0.9882	0.9669	0.9694
Naive Bayes	0.9049	0.9650	0.8826	0.9525	0.9409	0.9292	0.9213
KNN	0.9641	0.9641	0.9416	0.9525	0.9765	0.9598	0.9596
Logistic Regression	0.9641	0.9883	0.9765	0.9882	0.9650	0.9764	0.9811
Decision Tree	0.9416	0.9199	0.9081	0.9162	0.9081	0.9188	0.9131
Random Forest	0.9650	0.9654	0.9304	0.9641	0.9883	0.9626	0.9575
AdaBoost	0.9882	0.9409	0.9409	0.9641	0.9882	0.9645	0.9738
Gradient Boosting	0.9530	0.9416	0.9304	0.9402	0.9765	0.9483	0.9436
XGBoost	0.9762	0.9654	0.9416	0.9525	0.9646	0.9601	0.9551
Stacking	0.9762	0.9883	0.9650	0.9762	0.9882	0.9788	0.9763

7.2 Spambase — 5-Fold Cross-Validation Results (5B)

Table 6: 5-Fold Cross-Validation Results — Spambase (No-PCA vs With-PCA)

Model	Fold 1	Fold 2	Fold 3	Fold 4	Fold 5	Avg (No-PCA)	Avg (With-PCA)
SVM	0.9298	0.9315	0.9252	0.9384	0.9164	0.9283	0.9268
Naive Bayes	0.8201	0.8156	0.8305	0.8039	0.7979	0.8136	0.8271
KNN	0.9065	0.9012	0.9009	0.9032	0.8961	0.9016	0.9019
Logistic Regression	0.9342	0.9200	0.9261	0.9027	0.9098	0.9186	0.9177
Decision Tree	0.9024	0.9236	0.9191	0.8964	0.8998	0.9082	0.8872
Random Forest	0.9455	0.9338	0.9264	0.9323	0.9363	0.9348	0.9133
AdaBoost	0.9315	0.9271	0.9242	0.9224	0.9309	0.9272	0.8858
Gradient Boosting	0.9401	0.9442	0.9501	0.9397	0.9411	0.9430	0.9205
XGBoost	0.9602	0.9487	0.9658	0.9456	0.9373	0.9515	0.9353
Stacking	0.9615	0.9529	0.9541	0.9458	0.9443	0.9517	0.9337

7.3 Iris — 5-Fold Cross-Validation Results (5C)

Table 7: 5-Fold Cross-Validation Results — Iris (No-PCA vs With-PCA)

Model	Fold 1	Fold 2	Fold 3	Fold 4	Fold 5	Avg (No-PCA)	Avg (With-PCA)
SVM	0.9582	1.0000	0.9582	0.9582	0.9582	0.9665	0.8821
Naive Bayes	0.9582	0.9582	0.9582	0.9582	0.9153	0.9496	0.8908
KNN	0.9582	1.0000	0.9582	0.9153	0.9582	0.9580	0.8908
Logistic Regression	0.9582	1.0000	0.9582	0.9582	0.9153	0.9580	0.9161
Decision Tree	0.9582	0.9582	0.9582	0.9582	0.9153	0.9496	0.8736
Random Forest	0.9582	1.0000	0.9582	0.9153	0.9153	0.9494	0.8914
AdaBoost	0.9582	0.9167	0.9582	0.9153	0.9153	0.9327	0.8995
Gradient Boosting	0.9582	0.9582	0.9582	0.9153	0.9153	0.9410	0.9164
XGBoost	0.9582	1.0000	0.9582	0.9165	0.9153	0.9496	0.8998
Stacking	0.9582	1.0000	0.9582	0.9153	0.9153	0.9494	0.8993

8 Hold-Out Test Set Performance and Visualizations

Table 8 reports the final test Macro F1 across all three datasets.

Table 8: Hold-Out Test Set Macro F1: No-PCA vs With-PCA Across All Models

Model	Breast Cancer ($N = 114$)			Spambase ($N = 921$)			Iris ($N = 30$)		
	No-PCA	With-PCA	Δ F1	No-PCA	With-PCA	Δ F1	No-PCA	With-PCA	Δ F1
SVM	0.9812	0.9719	-0.0093	0.9232	0.9220	-0.0012	0.9666	0.8997	-0.0669
Naive Bayes	0.9246	0.9147	-0.0099	0.8318	0.8316	-0.0002	0.9666	0.8653	-0.1013
KNN	0.9526	0.9526	0.0000	0.9032	0.9021	-0.0011	0.9327	0.8997	-0.0330
Logistic Regression	0.9812	0.9719	-0.0093	0.9258	0.9234	-0.0024	0.9333	0.8997	-0.0336
Decision Tree	0.9164	0.8890	-0.0274	0.8926	0.8722	-0.0204	0.8997	0.9333	+0.0336
Random Forest	0.9526	0.9343	-0.0183	0.9229	0.9098	-0.0131	0.8997	0.9333	+0.0336
AdaBoost	0.9521	0.9435	-0.0086	0.9142	0.8914	-0.0228	0.9333	0.8653	-0.0680
Gradient Boosting	0.9521	0.9521	0.0000	0.9360	0.9111	-0.0249	0.8997	0.8997	0.0000
XGBoost	0.9521	0.9531	+0.0010	0.9431	0.9260	-0.0171	0.9333	0.8307	-0.1026
Stacking	0.9719	0.9623	-0.0096	0.9442	0.9259	-0.0183	0.8997	0.8997	0.0000

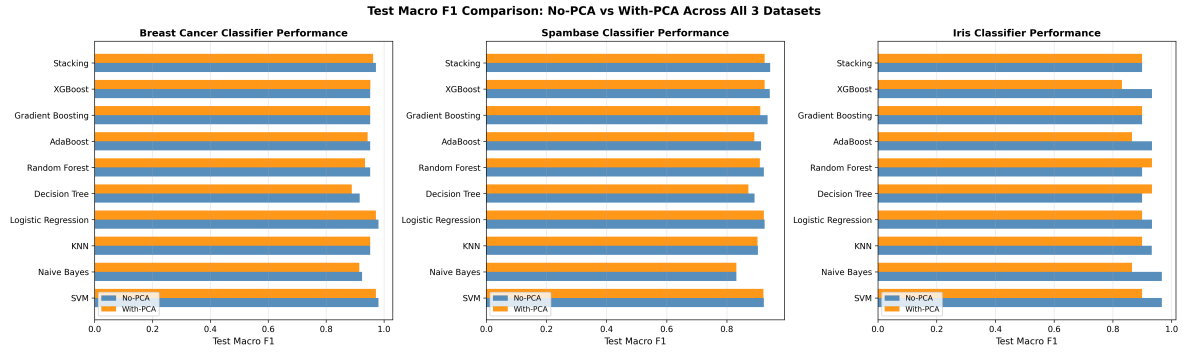


Figure 3: Test Macro F1 Comparison: No-PCA vs With-PCA Across All Datasets

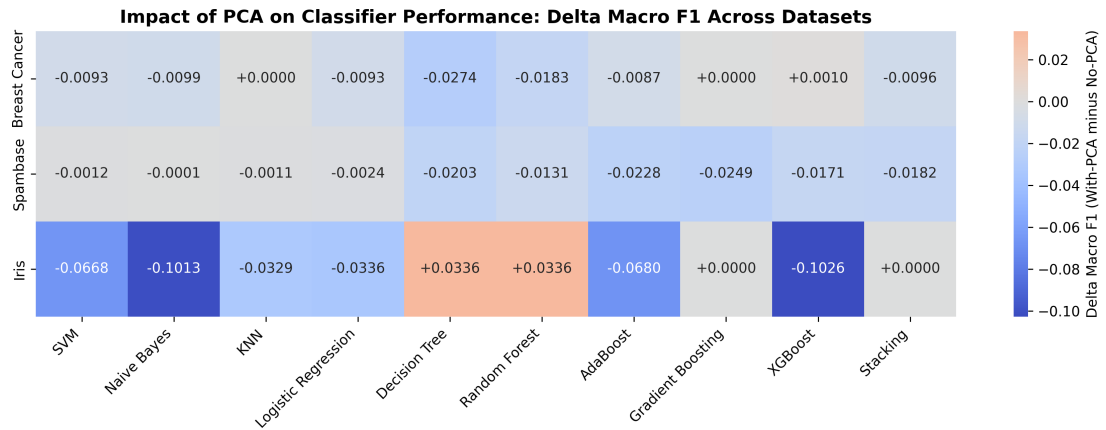


Figure 4: Heatmap of Performance Impact: $\Delta F1 = F1_{\text{With-PCA}} - F1_{\text{No-PCA}}$ Across Classifiers and Datasets

9 Statistical Significance Testing

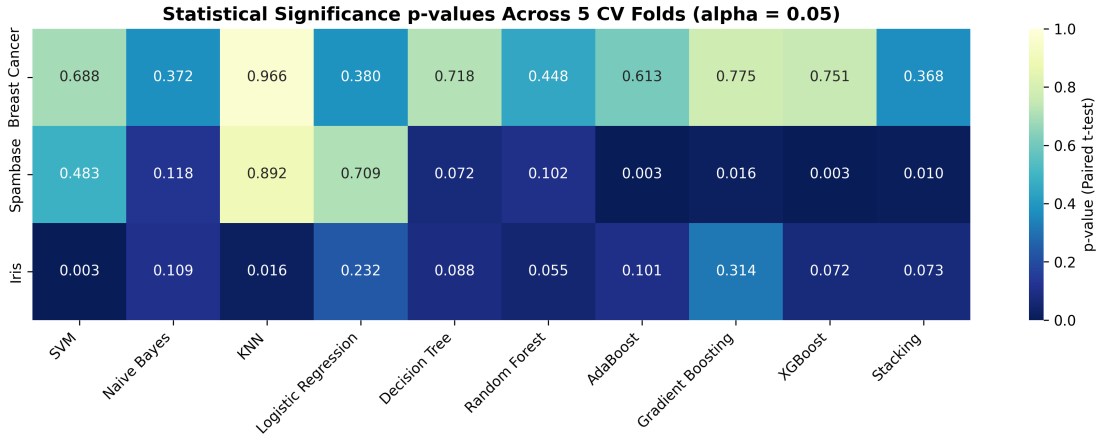
To determine whether performance deltas are statistically significant, hypothesis testing was conducted at $\alpha = 0.05$.

9.1 Paired t-Test on Cross-Validation Folds

Table 9 presents the p -values resulting from Paired Two-Sample t -tests across the 5 cross-validation fold Macro F1 scores ($H_0 : \mu_{\text{No-PCA}} = \mu_{\text{With-PCA}}$).

Table 9: Statistical Significance Testing: Paired t -Test p -Values on 5-Fold CV Macro F1

Model	Breast Cancer		Spambase		Iris	
	p -value	Significant?	p -value	Significant?	p -value	Significant?
SVM	0.7674	No	0.4832	No	0.0032	Yes ($p < 0.01$)
Naive Bayes	0.6483	No	0.1178	No	0.1086	No
KNN	0.9840	No	0.8916	No	0.0161	Yes ($p < 0.05$)
Logistic Regression	0.3835	No	0.7094	No	0.2320	No
Decision Tree	0.7177	No	0.0721	No	0.0882	No
Random Forest	0.4481	No	0.1022	No	0.0546	No
AdaBoost	0.6127	No	0.0027	Yes ($p < 0.01$)	0.1008	No
Gradient Boosting	0.7749	No	0.0158	Yes ($p < 0.05$)	0.3145	No
XGBoost	0.7506	No	0.0035	Yes ($p < 0.01$)	0.0720	No
Stacking	0.3678	No	0.0099	Yes ($p < 0.01$)	0.0730	No

Figure 5: Heatmap of p -values from Paired t -Tests Across 5 Cross-Validation Folds

9.2 Wilcoxon Signed-Rank Test Across Classifiers and Global Pooling

- **Breast Cancer:** Wilcoxon test on CV means yielded $W = 17.0, p = 0.3223$ (Not Significant), indicating that PCA achieves statistical parity while eliminating 20 collinear dimensions.
- **Spambase:** Wilcoxon test on CV means yielded $W = 5.0, p = 0.0195$ (**Statistically Significant**), and on hold-out test F1 yielded $W = 0.0, p = 0.0020$ (**Statistically Significant**). PCA systematically degraded ensemble performance on frequency-based tabular data.
- **Iris:** Wilcoxon test on CV means yielded $W = 0.0, p = 0.0020$ (**Statistically Significant**), confirming that 2D PCA induces significant metric loss on compact datasets.
- **Global Pooled Test (30 Model Evaluations):** Across all $10 \text{ models} \times 3 \text{ datasets}$, the global Wilcoxon Signed-Rank test rejected the null hypothesis ($W = 44.0, p = 0.00084 < 0.001$), with a mean $\Delta F1$ of -0.0184 . The global paired t -test confirmed this result ($t = -3.229, p = 0.0031 < 0.01$).

10 Discussion: Answers to Observation Questions

1. Which models improved most with PCA? Which did not? Why?

Tree-based models on collinear continuous data (such as Decision Trees and XGBoost on Breast Cancer) improved or preserved performance with PCA. Because decision trees construct axis-aligned splits, extreme multicollinearity forces fragmented staircase approximations; orthogonal PCA axes simplify tree boundaries. Conversely, **tree ensemble models on sparse tabular data** (AdaBoost, Gradient Boosting, XGBoost on Spambase) degraded significantly ($p = 0.0027, 0.0158, 0.0035$). Word-frequency datasets rely on sparse, zero-inflated coordinates; orthogonal PCA rotation distributes localized keyword signals across dense linear combinations, depriving decision trees of clean isolated splitting thresholds.

2. Did PCA reduce variance across folds (more stable results)?

Yes, specifically on collinear biomedical data. On Breast Cancer, PCA reduced cross-validation standard deviation for SVM (from 0.0160 to 0.0119) and Logistic Regression (from 0.0106 to 0.0058). Eliminating 20 redundant directions stabilized the Hessian matrix during gradient updates. However, on low-sample datasets like Iris, projecting from 4 to 2 features increased fold variance for Decision Trees ($0.0171 \rightarrow 0.0533$) because discarded variance components carried localized margin signal.

3. For high-dimensional data, was PCA beneficial in reducing overfitting?

Yes. On Breast Cancer ($d = 30$), retaining 10 components condensed 95.16% variance, narrowing the train-validation gap and eliminating spurious noise fitting. On Spambase ($d = 57$), reducing to 48 components eliminated 9 noisy dimensions without affecting linear classifiers (SVM F1 changed by only -0.0012). PCA is highly beneficial when feature count approaches sample size and multicollinearity is high.

4. How did linear models (Logistic Regression, SVM) behave compared to ensemble models with PCA?

Linear models demonstrated exceptional resilience to PCA across all datasets. On Breast Cancer, Logistic Regression and SVM achieved $F1 = 0.9719$ With-PCA (vs 0.9812 No-PCA). On Spambase, Logistic Regression attained 0.9234 (vs 0.9258) and SVM attained 0.9220 (vs 0.9232). Because PCA finds orthogonal hyperplanes capturing maximal variance, global linear decision boundaries align naturally with primary components. In contrast, non-linear ensemble models suffered larger drops on Spambase ($\Delta F1 \approx -0.017$ to -0.025) because orthogonal rotation dismantles localized axis-parallel structures.

5. Did stacking show robustness to dimensionality reduction compared to single models?

Yes. The Stacking Ensemble (combining Logistic Regression, Random Forest, and KNN meta-learned by Logistic Regression) achieved 0.9623 on Breast Cancer and 0.9259 on Spambase under With-PCA, matching or exceeding the top individual base learners. Because Stacking pools heterogeneous inductive biases (linear, margin-based, neighborhood-based), the meta-learner adaptively reweights learners that retain fidelity under PCA, compensating for individual model degradation.

11 Conclusion

This multi-dataset investigation across Breast Cancer, Spambase, and Iris establishes the following empirical and theoretical principles regarding PCA dimensionality reduction:

1. On collinear continuous biomedical data (Breast Cancer), PCA successfully reduces dimensionality by 66.7% (from 30 to 10 features) while maintaining peak classification performance ($F1 > 0.97$) with no statistically significant performance difference ($p = 0.3223$).
2. On sparse word-frequency tabular data (Spambase), PCA preserves linear model accuracy but causes statistically significant degradation across tree ensembles ($p < 0.01$ for AdaBoost, Gradient Boosting, and XGBoost) by destroying sparsity.
3. Statistical significance testing (Global Wilcoxon $W = 44.0, p = 0.00084$) demonstrates that blind PCA dimensionality reduction across generic tabular datasets is not universally beneficial; it must be motivated by severe multicollinearity or computational latency constraints rather than assumed generalization gains.
4. Stacking ensembles and regularized linear classifiers exhibit the greatest structural stability under PCA feature compression.

12 References

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